CASE STUDY: STUDENT RECORD MANAGEMENT SYSTEM

String-Based Implementation (Without Student Objects)

Objective

Develop a menu-driven Java application to manage student records using String-based data handling. Store each student's details as a single String record in a text file named **student.txt**.

Problem Statement

Create a Java program that allows users to add, view, search, delete, and update student records. The menu must continue to appear after each operation until the user selects Exit.

Student Record Details

Each record must contain these fields in the given order:

- Roll Number
- Name
- Age
- Marks in Subject 1
- Marks in Subject 2
- Marks in Subject 3
- Marks in Subject 4
- Marks in Subject 5
- Grade

Required Menu Options

Option	Operation	Description
1	Add Record	Accept student details and append the record to student.txt.
2	Read All Records	Read and display every student record in the file.
3	Read by Roll Number	Search for and display a student using the roll number.
4	Read by Name	Search for and display student record(s) using the name.
5	Delete by Roll Number	Delete the record matching the entered roll number.
6	Update by Roll Number	Find a student by roll number and update name, age, five marks, and grade. Keep the roll number u
7	Exit	Terminate the application.

Implementation Requirements

- Use Java String variables to represent complete student records.
- Separate fields with a delimiter such as a comma. Example: 101,Rahul,21,85,90,78,88,92,A
- Use split(",") or equivalent String operations to extract fields for searching and displaying records.
- Use ArrayList or another suitable collection to hold records temporarily during update and delete operations.
- Use FileWriter in append mode when adding a new record.
- For update and delete, read the records, modify the collection, and rewrite the records to student.txt.

- Validate user input, including numeric values and non-empty text fields.
- Close file and Scanner resources properly; try-with-resources is recommended.
- Do not create a Student class or Student objects. This version must use Strings for record handling.

Sample Record Format

```
101,Rahul,21,85,90,78,88,92,A
```

Expected Deliverables

- Java source code for the String-based implementation.
- A student.txt file containing sample records for testing.
- Successful execution of all seven menu options.

Note: This task is intentionally String-based. A separate version can later be implemented using a Student class and Student objects.